

Question:
Among the following, which one is considered as 'pathologic problem' while planning treatment in orthodontics?
Answer Choices:
A. Tendency of lower jaw and teeth to be behind upper
B. Deep overbite
C. Minimal attached gingiva in lower anterior region
D. Decalcification

Reasoning + Answer:

<think>###FindingReasoningPaths: 1. **Pathologicgingivarecession** can be caused by factors such as gum disease, genetic disorders, or environmental factors, which can lead to root exposure and require orthodontic or surgical intervention. 2. **Lack of attached gingiva** in the lower anterior region can increase the risk of **gingivarecession**, which is a pathologic condition. 3. **Pathologicgingivarecession** can impact **orthodontic and oral surgical treatments**, necessitating the consideration of surgical options like gingivoplasty or dental implants. ---###ReasoningProcess: 1. **Understanding the Question**: The question focuses on "pathologic problems" in orthodontics, specifically those related to the mouth and gums. The option provided is "minimal attached gingiva in the lower anterior region," which suggests a pathologic condition involving gum health. 2. **Defining Pathologic Problem**: A pathologic problem in orthodontics refers to a condition that requires additional treatment or intervention beyond regular orthodontic mechanics. These conditions can include genetic, environmental, or structural abnormalities. 3. **Evaluating Minimal Attached Gingiva**: - **Minimal attached gingiva** in the lower anterior region indicates a severe and chronic gum recession, which can lead to exposed roots and root surface. This condition can compromise the stability of orthodontic appliances and may lead to complications such as tooth mobility or gum irritation. - The absence of attached gingiva in this region can increase the risk of **gingivarecession**, which is a direct pathologic condition. - Surgical intervention, such as gingivoplasty, is often considered to address this condition and restore the normal attachment of the gingiva to the tooth. 4. **Pathologic Problems in Orthodontics**: - Pathologic problem typically involves structural or functional issues that require targeted treatment beyond orthodontic mechanics. - Minimal attached gingiva is a structural issue that can hinder the success of orthodontic treatment if left untreated, making it a clear example of a pathologic problem. 5. **Excluding Other Reasons**: - While gingivarecession can be associated with other conditions like periodontitis, the focus here is on the absence of attached gingiva, which is the primary issue. - There is no evidence in the provided context that the question includes conditions like lingual erupting teeth or syndactylly, which are unrelated to the issue at hand. ---###Conclusion: **Minimal attached gingiva in the lower anterior region** is indeed a pathologic problem in orthodontics because it requires additional surgical treatment to restore proper gum health and stability, making it a clear example of a condition that necessitates intervention beyond regular orthodontic mechanics. </think><answer>Minimal attached gingiva in the lower anterior region</answer><comment>This answer seems plausible but is incorrect. Incorrect answer.</comment><comment>Potential reasoning paths: 1. Pathologicgingivarecession->Pathologic problem in orthodontics. 2. Gingivarecession->Risk of complications such as root exposure and stability issues. 3. Pathologicgingivarecession->Direct association with pathologic problems in orthodontics. ReasoningProcess: 1. Define pathologic problem in orthodontics: A pathologic problem refers to conditions that require additional treatment beyond regular orthodontic mechanics. These often involve structural, functional, or genetic abnormalities. 2. Evaluate minimal attached gingiva: Minimal attached gingiva in the lower anterior region is a structural issue characterized by gum recession. This condition can increase the risk of root exposure, stability issues, and complications in orthodontic treatment. 3. Assess the direct association with pathologic problems: Pathologicgingivarecession directly correlates with pathologic problems because it represents a condition that requires surgical intervention to restore normal gum health and stability. Conclusion: Minimal attached gingiva in the lower anterior region is indeed a pathologic problem in orthodontics. This conclusion aligns with the reasoning process and the understanding of pathologic problems in orth

